Dwarak A

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.11
Vels Vidyashram	Class XII (CBSE)	2024	96.00%
Vels Vidyashram	Class X (CBSE)	2022	98.40%

SCHOLASTIC ACHIEVEMENTS

Dream11 FIFS Gamethon 1.0: Secured **2nd Place** among **5,000+ national participants** as part of a 7 member team, developing a high performance fantasy sports prediction model.

World Scholar's Cup 2019: Qualified for the Global Round, securing **Top 35%** in Literature and History, **Silver Medals** in Science and Special Area Challenges, and rankings of **35th in Team Bowl** and **45th in Writing Champions**.

POSITIONS OF RESPONSIBILITY

Design Core, Game Development Club, IIT Hyderabad:

Developed game-ready 3D character assets through modeling, rigging, and partial animation for a game created for Milan '26, IIT Hyderabad's annual inter-hostel festival.

PROJECTS

VLSI Global Routing Congestion Prediction using Deep Learning

- Engineered a congestion prediction pipeline using a **decoupled U-Net** with a **ResNet-34 backbone** to generate VLSI global routing congestion heatmaps from **macro placement**, **wire density**, and **pin density** features, enabling pre-routing design feedback for physical design closure.
- Designed a **Hotspot Focal Loss** function with **warmup based scheduling** to improve convergence on rare high utilization routing regions, stabilizing training against class imbalance.
- Architected the full ML pipeline, **data preprocessing, training, evaluation, and visualization**, supporting scratch, fine tuned, and frozen transfer learning configurations for benchmarking.
- Benchmarked against commercial EDA routing tools, reaching **0.877 Hotspot F1**, **0.0817 NRMSE**, and **0.484 SSIM** at **~40ms GPU inference latency** per layout.

Sports Analytics Prediction Model

- Built a player performance prediction model using **Random Forest** and **XGBoost** to forecast performance metrics from historical batting and bowling statistics.
- Processed multi-sheet match datasets using **Pandas**, merging and cleaning records to prepare structured features for model training.
- Developed an automated player ranking and selection system based on predicted performance scores, optimizing composite team output under constraint based selection rules.
- Containerized the full pipeline with **Docker** for reproducible deployment and consistent execution across environments.

Custom RISC-V ISA Extension for Hardware Accelerated FFT

- Extended the **open source Spike RISC-V ISA simulator** with custom instructions to offload FFT computation, removing the floating point hardware dependency for embedded signal processing.
- Implemented microarchitectural datapath logic for **CORDIC based twiddle factor generation**, **Q31 saturating multiplication**, **single cycle bit reversal**, and **packed 64 bit complex arithmetic** to cut butterfly operation instruction count.
- Delivered **5x to 6.5x execution speedup** over software baselines across FFT sizes from **64 to 4096 points**, validated at **70 to 90 dB SNR** against double precision references using Q31 fixed point math.

Multithreaded Gradient Computation for Linear Regression (Pthreads)

- Parallelized gradient computation across **10 threads** using **POSIX threads** in C++, cutting execution time from **245ms to 36ms** (**6.8x speedup**) on a **10,000 sample, 100 feature** synthetic dataset.
- Built a reproducible benchmarking framework to validate output correctness against single threaded baselines under correlated feature conditions.

SKILLS

Dev Tools and Platforms: Git, Docker, MySQL, Linux, VS Code, Neovim

Languages: Python, C++, C, SQL, Bash

Computer Architecture and Systems: Computer Architecture, Microarchitecture Design, Pipeline Optimization, Parallel Computing (POSIX Threads), Fixed Point Arithmetic, DSP Hardware Design

RELEVANT COURSES

Data Structures & Applications, Computer Architecture, Digital Systems, Introduction to Programming, Scientific Programming, Probability, Statistics, Machine Learning

EXTRACURRICULAR

National Sports Organization (NSO) - Badminton

Actively participated in the college's NSO badminton program, attending regular practice sessions and competing in intra-college badminton events.

Competitive Programming

Achieved a **Codeforces rating of 1233** and a **LeetCode rating of 1510**, solving algorithmic problems spanning data structures, graph theory, and dynamic programming.